# Lab 4: VibeCoding

Tic-Tac-Toe (Pygame): chat history with Claude

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**Name / SRN:** Aanya · PES2UG24CS011
**Repository:** SETAPESU26/09_tic_tac_toe
**LLM used:** Claude (chat)
**Tasks covered:** Task 1 (win/draw fixes), Task 3 (move validation), Task 2 (scoreboard), Task 4 (first-player choice, restart controls), plus cleanup, on-screen buttons and a final check

> **How to read this:** Blue-labelled blocks are my prompts and the other blocks are Claude's replies; both are reproduced as in the original chat export. Where the export pasted raw tool output (full file listings, "file updated" confirmations, image data), it is replaced here by a one-line grey note such as "Viewed file: game/rules.py". Spacing in my prompts was normalised and the numbered items were put on separate lines.

**PROMPT 1 · YOU**

> Here is my Tic-Tac-Toe pygame repo. Fix Task 1 from the README: diagonals must count as wins, a win on the last cell must be a win and not a draw, and no moves after the round ends.
> *Attached: 09_tic_tac_toe-main.zip (6.8 KB)*

- *Viewed file: README.md*
- *Viewed file: game/rules.py*
- *Viewed file: game/game_engine.py*
- *Viewed file: main.py*
- *Edited file: game/rules.py (updated successfully)*
- *Edited file: game/game_engine.py (updated successfully)*
- *Edited file: game/game_engine.py (updated successfully)*

**CLAUDE**

Quick headless test.

I fixed all three Task 1 problems, and a headless test confirmed each one. I haven't played the game in a window.

- **Diagonals:** `game/rules.py` now checks both diagonals along with the rows and columns, so all 8 winning lines are detected.
- **Win on the last cell:** in `game/game_engine.py`, `check_round_end` now looks for a winner before checking for a full board. A winning move that fills the last cell is reported as a win, and a draw only happens when the board is full with no winner.
- **No moves after the round ends:** `handle_click` now returns straight away once the round is over.

The test used a diagonal win for X and one for O. It also used a full board where X wins on the last cell, a full board that is a real draw, and a click after the round ended, which left the board unchanged.

I left the occupied-cell overwrite alone, since that is Task 3.

The fixed files are in `/home/claude/ttt/09_tic_tac_toe-main/tic-tac-toe/`.

**PROMPT 2 · YOU**

> Task 3: reject clicks on occupied cells entirely, so the board and turn stay unchanged.

- *Edited file: game/game_engine.py (updated successfully)*

**CLAUDE**

My test click landed above the board (it starts at y=100), so that check proved nothing. Rerunning it on the actual cell.

Task 3 is fixed. Clicking an occupied cell now does nothing: the board stays as it was and it is still your turn (X).

The change is in `handle_click` in `game/game_engine.py`, which now returns early if the clicked cell isn't empty. A headless test confirmed it: clicking a cell holding an O left the board and turn unchanged, and clicking an empty cell still places an X.

**PROMPT 3 · YOU**

> Task 2: add a persistent scoreboard for X wins, O wins and draws. It must survive a round restart and clear only on a full match reset.

- *Viewed file: game/renderer.py*
- *Viewed file: game/game_engine.py (partial)*
- *Image output (screenshot)*

**CLAUDE**

The scoreboard is in. It shows X wins, O wins and draws at the top of the window, and it stays put when you start a new round. I checked this with a headless test and a screenshot of the layout; I haven't played it live.

- **Counting:** each finished round adds one to X, O or Draws, and only once. Clicking after the round ends can't double-count.
- **Controls:** R now starts a new round and keeps the scores. Before, R wiped everything. A new key, M, resets the whole match and clears the scoreboard. There's a small hint line under the scores so both keys are visible.
- **Task 4:** this gives you the two separate restart controls that Task 4 asks for. Task 4's first-player choice is still not done.

The changes are in `game/game_engine.py`: `new_round()`, `reset_match()` and the scoreboard drawing.

**PROMPT 4 · YOU**

> Task 4: add first-player choice (X or O starts, applying from the next round) and two separate controls: restart round, and reset match

- *Image output (screenshot)*
- *Tool error: file does not exist*

**CLAUDE**

My one-liner truncated `renderer.py` before reading it. I'm restoring it from the earlier read, with the new height.

- *Wrote file: game/renderer.py*
- *Image output (screenshot)*

Task 4 is done: you can now choose who starts, and there are two separate restart controls. I tested the logic headless and checked the layout in a screenshot, but I haven't played it live.

- **F** switches who starts, X or O. The current round isn't touched. The choice applies when the next round begins, and the screen shows "Next round: X/O starts". If O starts, the computer plays the opening move straight away.
- **R** restarts the round and keeps the scoreboard. The new round uses the current starter choice.
- **M** resets the match and clears the scores. It also keeps your starter choice.

I also made a few layout changes:

- The window is taller (600 instead of 500), because the win/draw banner was half cut off at the bottom edge.
- The key hints are now at the bottom.
- The banner now reads "X wins! R: new round" so it fits the width.

While editing, I accidentally emptied `game/renderer.py` and rewrote it with its original contents plus the new window height. The game runs again and the screenshot renders correctly. Compare it against your original zip if you want to be sure nothing differs.

The changes are in `game/game_engine.py` and `game/renderer.py`, in `/home/claude/ttt/09_tic_tac_toe-main/tic-tac-toe/`.

**PROMPT 5 · YOU**

> provide the final zip file

**CLAUDE**

The final zip is `09_tic_tac_toe-final.zip`. It has the same folder layout as your original and includes all four tasks.

Controls in the game: click to place X, **R** restarts the round, **M** resets the match and scores, and **F** switches who starts next round.

**PROMPT 6 · YOU**

> Two cleanups, no behavior changes:
> 1. The docstring at the top of game/game_engine.py still says "Starter version: no scoreboard yet, no first-player choice…" . Rewrite it to describe the current game: scoreboard, move validation, first-player choice, and the R / M / F controls.
> 2. Add a short "Controls" section to README.md: click to play X, R restarts the round and keeps the score, M resets the match and clears the score, F chooses who starts the next round. Give me the full updated files.

▸ *Viewed file: game/game_engine.py*

**CLAUDE**

Both cleanups are done, and I sent the full `game_engine.py` and `README.md`. I also re-zipped the project as `09_tic_tac_toe-final.zip`.

- **`game_engine.py`:** the "Starter version…" paragraph is replaced with a description of the current game. It covers the scoreboard, win/draw detection, move validation and the first-player choice, plus the Click / R / M / F controls. Only the docstring changed, and a quick run of the game logic still works.
- **`README.md`:** there's a new "Controls" section right after "Getting Started", with the four controls you listed.

The README's existing "Controls:" line under Setup still says "Press R for a new round". It isn't wrong, but it's now partly redundant with the new section. I left it alone since you asked for no other changes.

**PROMPT 7 · YOU**

> Add three clickable on-screen buttons below the board: "New Round (R)" , "Reset Match (M)" and "First: X/O (F)" . They should do the same as the R, M and F keys. Keep the keyboard shortcuts working. The window is 400x600, so make sure the buttons and the existing text don't overlap. Only change renderer.py, game_engine.py and main.py if needed, and give me the full files.

▸ *Viewed file: game/renderer.py*

▸ *Viewed file: game/game_engine.py*

▸ *Image output (screenshot)*

**CLAUDE**

Layout is clean with no overlaps. Re-zipping and sending the files.

The three buttons are in, and the R, M and F keys still work. I tested the clicks and keys headless and checked the layout in a screenshot, but I haven't clicked them in a real window. I've sent the full `renderer.py` and `game_engine.py`, plus a refreshed zip. `main.py` needed no change.

- **Buttons:** “New Round (R)” and “Reset Match (M)” sit side by side. “First: X (F)” sits below them and shows who starts next, switching to “First: O (F)” when toggled. They match the keys exactly: New Round keeps the score, Reset Match clears it, and First applies from the next round.
- **After a round ends:** the buttons still work, while clicks on the board stay blocked.
- **Layout:** the buttons sit at the bottom of the 400x600 window, below the win/draw banner. A test confirmed they don’t overlap each other or the banner and stay inside the window. The old key-hint text lines are gone, since the button labels now show the keys.
- **Hover:** a button shades slightly when the mouse is over it.

The README still describes the keys only. Add a line about the buttons if you want it mentioned there.

**PROMPT 8 · YOU**

> Re-read all the final code and check it against the README’s “Expected Behavior” list, point by point. Tell me if anything is missing or wrong.

*Viewed file: game/game_engine.py*

*Viewed file: game/rules.py*

*Viewed file: game/ai.py*

**CLAUDE**

I read the code; now I’m running an exhaustive check against each Expected Behavior line.

All six “Expected Behavior” points pass, and I found one visual bug outside that list. I read the code and ran checks, but I haven’t played it in a window.

**Expected Behavior, point by point**

- **All 8 winning lines (3 rows, 3 columns, both diagonals):** pass. Each line was tested for both X and O.
- **A win on the last cell scores as a win, not a draw:** pass. I checked every board that can occur at the end of a game: 110 are wins and 16 are draws, and each was scored correctly.
- **Clicks do nothing after the round ends:** pass. Across 5,000 full games, every board click after the end left the board, turn, score and winner unchanged.
- **Occupied-cell clicks are rejected, with board and turn unchanged:** pass. The same 5,000 games mixed in random clicks on occupied cells, and none changed anything.
- **The scoreboard survives a round restart and resets to zero on a full match reset:** pass. After 5,000 rounds the three counts added up to exactly 5,000, with one point per round and none double-counted. R and the New Round button kept the scores, and M cleared them.
- **The first-player choice takes effect on the next round:** pass. Pressing F mid-round changes nothing on the board. On the next round, O starting means the computer opens, and the turn then passes to you.

**One thing that’s wrong**

- **Misleading turn label:** once a round is over, the top line still shows a turn label that doesn’t match the result. After you win it says “Computer’s turn (O)”, and after the computer wins it says “Your turn (X)”. This was already in the starter code. It’s cosmetic and doesn’t affect scoring or moves. A fix would hide the turn label when the round is over, and I haven’t made it.

**Minor notes**

- The README’s Controls section doesn’t mention the on-screen buttons.
- The README’s older “Controls” line under Setup also still describes only R.
- You didn’t ask for a change to either, so I left them.

Want me to fix the turn label and add the buttons to the README?

**PROMPT 9 · YOU**

> Yes, please fix both:
> 1. When the round is over, replace the turn label with “Round over” (or hide it), so it never shows a turn that doesn’t match the result.
> 2. Update README.md: in the Setup section’s Controls line and in the Controls section, describe all controls: click to play X, R / “New Round” restarts the round and keeps the score, M / “Reset Match” clears the score, F / “First: X|O” chooses who starts the next round, and the on-screen buttons do the same as the keys. Give me the full updated files, and tell me what you changed.

▸ *Image output (screenshot)*

**CLAUDE**

Both fixes are in. I’ve sent the full `game_engine.py` and `README.md`, plus a refreshed `09_tic_tac_toe-final.zip`. A screenshot of a finished round confirmed the label change; I haven’t played it live.

**`game/game_engine.py`**

- When a round is over, the top-left label now reads “Round over”. While the round is running it still shows “Your turn (X)” or “Computer’s turn (O)”. That was the only change; the game logic is untouched.

**`README.md`**

- The Controls line in the Setup section no longer says only “Press R”. It now points to R / M / F, or the on-screen buttons, and refers to the Controls section below.
- The Controls section lists every control:
  - Click to play X.
  - R / “New Round” restarts the round and keeps the score.
  - M / “Reset Match” clears the score.
  - F / “First: X|O” chooses who starts the next round, and the button shows the current choice.
- It also says the buttons do the same as the keys and that the shortcuts still work.